MIL-B-36633
3 February 1970

MILITARY SPECIFICATION

BONE SURGERY SET, EXTENSION AND TRACTION, PIN TYPE, STEINMANN

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a bone surgery set suitable for surgical procedures involving traction reduction of fractures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
MMM-A-250	Adhesive, Water-Resistant, (for Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-36626	Pin, Bone

FSC 6515

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)

1908 TRACTOR, BONE PIN
1909 HANDLE, BONE PIN, STEINMANN

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The chuck handle and bone pin tractors shall be fabricated from the following materials in accordance with GG-I-526:

Component	GG-I-526 Class	GG-I-526 Type
Handle (excluding Jacobs Chuck)	3	304
Jacobs Chuck	4	416
Tractor, Bone Pin	3	304
Bone Pins	See MIL-P-36626, dated 6 January 1970.	

3.2 Hardness. After proper stress relieving Rockwell hardness of class 3, type 304 material, specified in 3.1, shall be B75 to B85 and C33 to C41 for class 4 material when tested as specified in GG-I-526.

3.3 Finish. Finish of the tractors shall be mirror or satin and finish of the chuck handle (including Jacobs Chuck) shall be mirror. The mirror and satin finish shall be as described in GG-I-526.

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3.3.1 Passivation. All components of the bone surgery set shall be properly passivated after completion of all final finishing operations.

3.4 Style, design and dimensions. Shall be a bone surgery set suitable for use in surgical procedures involving tractor reduction of fractures using the Steinmann procedures. The bone surgery set shall be furnished complete with one each of the following components:

3.4.1 Chuck handle. The chuck handle shall be fabricated from material specified in 3.1. The chuck shall be of the Jacobs design, having 3 jaws and capacity for 1/4 inch diameter rods. Shall be suitable for use with Steinmann bone pins with triangular shank ends. Shall be in accordance with DPSC Drawing 1909.

3.4.2 Tractors. Bone pin tractors shall be fabricated from material specified in 3.1. Tractors shall be of the Bohler design, "U" shaped, suitable for attachment to Steinmann bone pins. Tractors shall be medium and small size and one of each size shall be included with the bone surgery set. Shall be in accordance with DPSC Drawing 1908 for appropriate size and item identification. When tractors are furnished separately they shall be identified by the following stock numbers:

Reference Stock Number	Item Identification
6515-310-7840	TRACTOR, BONE PIN, Medium, 8-1/2 inch
6515-310-7860	TRACTOR, BONE PIN, Small, 5 inch

3.4.3 Bone pins. Bone pins shall be of the following sizes and styles of Military Specification MIL-P-36626, dated 6 January 1970:

Size	Style	Reference Stock Number
2	2	6515-310-7900
3	2	6515-310-7920
4	3	6515-310-7940

One of each of the above sizes shall be furnished with the bone surgery set.

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3.5 Resistance to boil test and copper sulfate test. All components of the bone surgery set shall show no evidence of corrosion when subjected to the boil test as specified in 4.4.1. In addition, the Jacobs chuck of the handle shall show no evidence of corrosion when subjected to the copper sulfate test as specified in 4.4.2.

3.6 Identification marking. Bone surgery sets made in U.S.A. shall be marked with the name or registered trademark of the contractor and the term "stainless". Sets of foreign (other than U.S.) manufacture shall, in addition to the above, indicate the country of origin.

3.7 Workmanship. Workmanship shall be first class throughout. Bone surgery sets shall be free from defects which detract from their appearance or impair their serviceability, proper functioning, and intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the austenitic and martensitic stainless steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table I.

Table I. Sampling Criteria for Examination

Type defect	Inspection Level*	AQL (percent defective)
Major	S-4	1.0
Minor	S-2	2.5

* If acceptance number is not exceeded, lot shall be accepted. If rejection number is equalled or exceeded, then the inspection at level II may be performed to determine the final rejection or acceptance.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II.

Table II. Sampling Criteria for Tests

Component	Characteristic	Req't	Test Procedure	Unit of Product	Insp Level	AQL
Bone Surgery Set	Dimensions	3.4	---	1-Set	S-2	*
Jacobs Chuck	Copper Sulfate Test	3.5	4.4.2	1-Chuck	S-2	*
Handle and Tractors (including Jacobs Chuck)	Boil Test	3.5	4.4.1	1-handle & 1-tractor	S-2	*
	Hardness	3.2	GG-I-526	1-handle & 1-tractor	S-2	*
Bone Pins	ALL REQUIREMENTS SEE MIL-P-36626					

* Acceptance number shall be zero for all sample sizes.

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4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in the construction of the bone surgery set shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The bone surgery sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Inspection, however, shall not be restricted to defects listed in Table III. Any bone surgery sets in the sample containing one or more defects shall be rejected and if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	None defined
Major	
101	One or more components of set missing.
102	Components not of finish specified.
103	Bone pins and tractors not of sizes specified.
104	Components not free from cracks, pits, scale, slivers, and tool marks.
105	Chuck not capable of accepting rods up to and including 1/4 inch diameter.
106	Chuck not suitable for use with Steinmann bone pins.
107	Tractors not suitable for use with Steinmann bone pins.
Minor	
201	Bone surgery sets not identified as specified in 3.6.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Boil test. All components of the bone surgery set shall be subjected to the boil test as described and specified in GG-I-526.

4.4.2 Copper Sulfate test. Jacobs chuck shall be subjected to the copper sulfate test as described in GG-I-526.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One bone surgery set or one tractor, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Item	Unit package	Intermediate package	Exterior container
Bone Surgery Set	1 unit	Not required	36 units
Tractor	1 unit	6 units	72 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package.

5.1.3.1.1 Bone surgery set. Each component comprising the set shall be thoroughly cleaned and wrapped in neutral tissue or other suitable anticorrosive material . Each set shall be further packaged in accordance with MIL-P-116, method IA-15. Container shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents.

5.1.3.1.2 Tractor. Each unit shall be thoroughly cleaned and wrapped in neutral tissue or other suitable anticorrosive material . Each unit shall be further packaged in accordance with MIL-P-116, method IA-15.

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5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Unit shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The bone surgery sets or their individual components are intended for use in Steinmann surgical procedures involving traction reduction of fractures.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
*b. Applicable Federal stock number (see 6.3).
c. Applicable levels of packing and packaging.

*NOTE: For Federal Stock Numbers of components see 3.4.2 and 3.4.3.

6.3 This specification covers the following item appearing in the Federal Supply Catalog as indicated below:

Federal Stock No.	Item Identification
6515-310-7800	BONE SURGERY SET, Extension and Traction, Pin Type, Steinmann

6.4 Supersession data. This specification includes requirements of Purchase Description No. 5, dated 24 November 1967 for FSN 6515-310-7800 and for Purchase Description No. 4 dated 12 May 1964 for components FSN 6515-310-7840 and 6515-310-7860.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1116

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-B-36633 FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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